United International University

School of Science and Engineering

Mid Term Examination; Year 2025; Trimester: Spring

Course: PHY 2105; Title: Physics; Sec: A-T

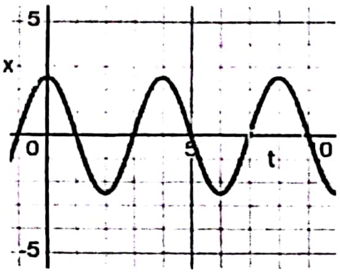
Full Marks: 30, Time: 1 Hour 30 Minutes

Any examinee found adopting unfair means will be expelled as per UIU disciplinary rules.

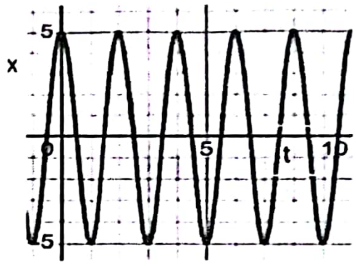
Questions no 1, 2, 3 are mandatory to answer. Answer anyone from question no 4 and 5.

1.(a) The drawing shows plots of the displacement x versus the time t for two objects undergoing simple harmonic motion. Which object, I, II, has the greatest maximum velocity? Justify your answer using necessary equations. — 2.5 CO1

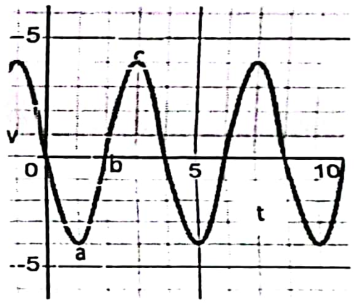
Object I



Object II



(b) Velocity vs. time graph is shown in adjacent Fig. What is the displacement at point a, what is acceleration at point b. at which position or positions where the maximum velocities occur? — 2.5 CO1



(c) Draw a transverse wave and show the wavelength on the wave. — 1 CO1

2.(a) A mass of 500gm is hung from a spring extension is found 10cm. An additional mass 500gm mass is added to the spring and allow to oscillate. Calculate (i) Angular frequency and (ii) time period of the oscillation — 3 CO3

(b) A block attached to a spring is suspended vertically. If the block is pushed 10 cm upward from the equilibrium position and released at t = 0. The mass of the block is 2.5 kg and the spring constant is k = 25 N/m. — 3 CO3

(i) Calculate the potential (PE) energy at x = 3 cm.

(ii) Find out the position where KE=PE.

(iii) Calculate the total energy of the system.

3.(a) A technician constructed an RLC circuit with the value, C = 0.009 μF, L = 0.5 mH, and R= 200Ω respectively. — 3 CO3

(i) Whether the circuit is oscillatory, calculate the frequency of oscillation of the RLC circuit.

(ii) What will be the value of resistance R for which circuit will be over damping?

(b) For a damped oscillator, $m = 0.30$ kg, $k = 19.6$ N/m, and $b = 100$gm/s. The oscillator is released at $t = 0$ and the amplitude is 10 cm. | 4 | CO3

(i) Calculate the frequency of oscillations of the oscillator.

(ii) How long does it take for the amplitude of the damped oscillator to drop to one half of its initial value?

(c) Equation of a Simple Harmonic Wave is $y = 10 sin\pi(0.25x - 90t)$ | 3 | CO3

(i) Find out wave length of the wave,

(ii) Find out wave speed of the wave and

(iii) Frequency of the particles of medium.

4.(a) Derive the differential equation for Simple Harmonic Motion(SHM). Show that $x = Acos(\omega t + \phi)$ is the solution of the equation. | 4 | CO2

(b) For a mass-spring system oscillating in SHM, the equation of displacement is, $x = Asin\omega t$. Show that the total energy of the oscillator is, $E = \frac{1}{2}KA^2$. Plot energy vs. displacement graph. | 4 | CO2

5.(a) Show that, the equation of displacement of the particles of a medium for a progressive wave is $y = A\ sin\frac{2\pi}{\lambda}(vt\text{-}x)$. Calculate the value of $\frac{d^2y}{dt^2}$. Here the symbols have their usual meanings. | 4 | CO2

(b) Derive the differential equation of a mass-spring system oscillating in DHM. Write the conditions of three types of damped harmonic motion and graphically represent them by plotting displacement vs. time graphs. | 4 | CO2

CO1: Define different physical quantities with examples **CO2:** Derive/Show the various equations of SHM, DHM, wave motion etc.
CO3: Evaluate different numerical problems based on the basic characteristics of SHM, DHM.

$\gamma = \frac{b}{2\sqrt{m}}$

$v = \lambda f$

$\lambda = 1/\gamma$

$\lambda = \frac{2\pi}{K}$